

Simulation report

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Report of the simulation. Here are the main parameters. Main result:
Indeed, this protocol actually simulates polymeric gels.

Potential energy analysis

Structure factor analysis

Three length scales can be defined for the structural factor to analyze the system's structure. The large length scale regime is characterized by the condition $10^{-1.5} \leq |\vec{q}| \leq 10^{-0.5}$. An intermediate length scale regime is characterized by the criterion $10^{-1/2} \leq |\vec{q}| \leq 10^{1/2}$. A small length scale regime is described as $10^{-0.5} \leq |\vec{q}| \leq 10^{1.0}$. The three regimes are distinguished by gray, blue, and red, respectively.

Beginning with the structure factor of the initial condition, top panel, at the large length scale regime, it is observed that at low packing fractions, the intensity approaches one, whereas an increase in packing fraction leads to a reduction in intensity. The results indicates that the patchy particles at a low packing fraction are randomly and uniformly dispersed throughout the simulated box, as expected. Nonetheless, as the packing percentage increases, large scale density variations gradually start to be suppressed. In the intermediate length regime, the intensity of the structure factor for the system with $\phi > 1$ increases towards one, while the intensity of the structure factor for the system with the lowest packing fraction remains about at 1. This indicates that at these length scales, the initial state for systems with $\phi > 1$ begins to approximate that of an ideal gas, similar to the system with $\phi = 1$. Furthermore, at the small length scale regime, the intensity of all structure factors approaches 1. This signifies that the cores of the central particles are randomly distributed within the volume of the simulation box, which displays identical initial conditions to the minimum packing percentage.

Pore analysis